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ABERDARE TECHNICAL COLLEGE

ICT TECHNICIAN LEVEL 5

UNIT: COMPUTER SOFTWARE

ISCED UNIT CODE: 0619 451 06A

CONTINUOUS ASSESSMENT TEST (CAT) 3

ELEMENT: SOFTWARE MAINTENANCE, MONITORING, UPGRADES AND SAFETY PROCEDURES

TIME: 1½ HOURS

TOTAL MARKS: ~~50~~ 60

CANDIDATE NAME: Ruon Maine

REGISTRATION NUMBER: _____

DATE: 30/6/26

INSTRUCTIONS TO CANDIDATES

1. Answer ALL questions.
2. Answer sheets will be provided separately.
3. Read each question carefully before answering.
4. Marks for each question are indicated in brackets.
5. The total marks for this paper are 50.

1. Define the term software maintenance. (2 Marks)
2. State two importance of software maintenance. (2 Marks)
3. List two activities that may be included in a software maintenance schedule. (2 Marks)
4. Differentiate between:

a) Corrective maintenance

b) Preventive maintenance (4 Marks)

5. Match the following software maintenance types with their descriptions: (4 Marks)

a) Adaptive Maintenance

b) Perfective Maintenance

c) Preventive Maintenance

d) Corrective Maintenance

Descriptions:

i. Improving software performance and usability

ii. Correcting software faults and errors

iii. Preventing future software failures

iv. Modifying software to operate in a new environment

6. State any three types of software updates. (3 Marks)
7. State three Personal Protective Equipment (PPE) used when handling computer equipment. (3 Marks)
8. Explain four importance of software upgrades. (4 Marks)
9. Describe three types of software upgrades. (6 Marks)
10. Define software functionality monitoring. (2 Marks)
11. Explain the following types of operating system event logs: (8 Marks)

a) Error Event Logs

b) Warning Event Logs

c) Information Event Logs

d) Success Audit Event Logs

12. Prepare a simple monthly software maintenance schedule for a computer laboratory showing four maintenance activities. (4 Marks)
13. Explain three importance of software updates. (6 Marks)
14. Explain three safety measures observed during software maintenance activities. (6 Marks)
15. Explain the importance of proper use of tools and equipment when carrying out software maintenance. (6 Marks)

1] Software Maintenance → is the Process of modification, updating, and improving Software after it has been installed to fix errors, improve Performance, or adapt to it to new Requirements.

2] importance:

Fix Software Bugs and Errors.
Improves Software Performance and Security.

3] activities included in Software maintenance:

- Running antivirus Scan.
- installing Software updates and patches.

4] Differentiate b/w:

a] Corrective maintenance:

→ Fixes Software Errors and bugs after their use.

Preventive maintenance:

- Prevents future Software Failures by improving Software problems occur.

5] mention the Software maintenance types:

a] Adaptive maintenance → modification Software to work in a new Environment.

b] Protective Maintenance → improving Software Performance and reliability.

c] Preventive Maintenance → Preventing future Software Failures.

d] Corrective Maintenance → Correcting Software faults or errors.

Types of Software Updates

6] Security updates ✓

Feature updates ✓

- Bug fix updates ✓

7] Protective Equipments •

- Safety gloves ✓

Safety goggles ✓

Anti-static wrist strap ✓

5] Importance of Software updates.

Add new Feature ✓

Improve System Performance ✓

Fix Software Bugs ✓

Improve Computer Security ✓

9] 3 types of Software Upgrades.

- Firmware upgrade → Upgrading firmware to improve hardware performance ✓

- Application Software upgrade → Installing a new version of an application. ✓

- Operating System upgrade → Upgrading the OS to a new version. ✓

10] Software Functionality Monitoring

Checking that software operates correctly and performs as expected. → Process of ✓

11] Explain.

a) Error Event Logs → Record Serious Problems That Cause Software or Hardware Failures ✓

b) Warning Event Logs → Record Potential Problems That Lead to Future Failures ✓

- 1] Information Event Log → Record normal system activities such as Startup, Shutdown and Successful logins.
- 2] Success Audit Event Log
Records normal Successful logins such as Successful user logins.

12] Schedule

Week	Activity
Week 1	Update Operating System and Applications.
Week 2	Run Antivirus Scan
Week 3	Remove unnecessary Files & Software.
Week 4	Back up Important data and Check System P

1.

13] Importance of Software updates.

- Fixes bugs and errors.
- Protect the Computer from Security +
- Improves Performance and Compatibility with new hardware.

14] Safety Measures

- Use Licensed and trusted Software only
- Back up Important data before making
- Ensure Stable Power Conn.

15 Importance of Proper use of tools and



Equipment

- Prevents damage to Computer Components
- Improves Safety of the Technology
- Ensures Maintenance to Avoid Malfunction and Downtime
- Extends the lifespan of Computer Equipment

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